

Gürkan Doğukan Karakurt

GRADUATE RESEARCHER · CIVIL ENGINEER · SOFTWARE DEVELOPER

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MSc candidate and Research Assistant in Construction Engineering & Management at METU, supervised by Prof. Dr. Rifat Sönmez. Thesis applies graph neural networks to BIM models to generate complete construction precedence schedules from a building's geometry and standardized activity classifications. Background in civil engineering (BSc, METU) with a minor in Corporate Finance and a long-running practice in full-stack software development and graphic design.

EDUCATION

- 2025 – Present **MSc, Construction Engineering & Management**
Middle East Technical University (METU), Ankara
Research Assistant. Thesis on BIM-driven scheduling with graph neural networks, supervised by Prof. Dr. Rifat Sönmez. Transferred from the Geotechnical Engineering programme in Spring 2025.
- 2017 – 2023 **BSc, Civil Engineering**
Middle East Technical University (METU) — GPA 3.35 / 4.00
Coursework in structural analysis, geotechnics, hydraulics, construction management, and Building Information Modeling.
- 2019 – 2023 **Minor, Corporate Finance**
Middle East Technical University (METU) — GPA 3.39 / 4.00
Minor programme covering financial analysis, valuation, and quantitative methods alongside the civil engineering degree.

RESEARCH

Predicting Construction Precedence Relationships from BIM Geometry with Machine Learning

I am building a graph neural network that reads a BIM model and outputs the full construction schedule — every FS, SS, FF, SF precedence link — so that schedules regenerate themselves whenever the design changes.

BIM and the construction schedule still do not talk to each other; planners rebuild precedence by hand for every project. The model treats the BIM as a graph: nodes are building elements with their IFC class, geometry, and standardized activity codes (Unifomat II, OmniClass T22); edges encode topology like adjacency, support, and containment. Message passing across that graph scores every candidate precedence link, replacing free-form activity text with standardized vocabularies.

To my knowledge this is the first end-to-end whole-schedule precedence predictor grounded directly in BIM topology rather than free-form activity text, designed to transfer across firms and projects.

EXPERIENCE

Oct 2025 – Present	Research Assistant — Construction Engineering & Management Middle East Technical University (METU), Department of Civil Engineering Graduate research and teaching support within the Construction Engineering and Management division. Focus areas include BIM, graph neural networks for construction scheduling, and machine learning applied to the built environment.
Dec 2024 – Oct 2025	Software Engineer Mega Mühendislik, Türkiye Developed software for engineering workflows, contributing across the stack to ship reliable internal tooling before moving full-time into graduate research at METU.
2023 – Present	Founder & Full-Stack Developer — Diyet Cebimde Self-initiated · AI-assisted diet & nutrition platform Solo-built mobile platform that generates AI-driven weekly diet plans from each user's profile and goals, with photo-based meal recognition, manual tracking, weight/measurement logging, and a memory-equipped chatbot. Hybrid pipeline (Gemini for meal selection, PuLP linear programming for portion optimisation) and snapshot architecture keep historical plans stable across catalog updates. Three repositories: Django + DRF + PostgreSQL backend, React 18 + Vite admin panel, Expo + React Native mobile client.
2023 – 2024	Head of Quantitative Strategies Vortex · Remote (Georgia) Designed and developed algorithmic trading systems for financial markets — scalping and swing strategies, performance tuning, and per-investor mandate customisation.
Summer 2022	Civil Engineering Intern ES Group — Erdemir Port Project, Zonguldak Worked across site operations, planning, and progress payments on a port construction project. Contributed to steel-pile design and took responsibility in cost control and project scheduling.
2020 – 2024	Freelance Graphic Designer · 2024 Local Elections Lead Designer Independent Delivered posters, brochures, social media content, and campaign materials across freelance projects. Served as lead designer for a 2024 Turkish mayoral campaign, owning the full digital content production pipeline.

SKILLS

Research & Engineering: BIM (Revit, IFC), Construction Management (scheduling, CPM), Machine Learning (PyTorch, GNN (GraphSAGE / GAT)), Optimization (Linear Programming (PuLP)), AutoCAD (upper-intermediate), EPANET (advanced), SAP2000 (beginner-intermediate), MS Project (intermediate), MATLAB / Visual Basic (advanced)	Backend & Data: Python (advanced), Django + Django REST Framework (advanced), PostgreSQL (ArrayField, JSONField), JWT (SimpleJWT) + Google OAuth (production auth), Google Gemini API (production integration), Unicorn (production deployment)
Frontend & Mobile: TypeScript (advanced), React (advanced), Next.js (App Router) (advanced), React Native + Expo (advanced (EAS Build)), Tailwind CSS + NativeWind (advanced), React Query + Zustand (state management), Vite, Vercel, Railway (build & deploy), HTML / CSS / JavaScript (advanced)	Design: Photoshop (advanced), Illustrator (advanced), Premiere Pro (advanced), After Effects (intermediate)
Languages: Turkish (native), English (C1)	

Personal contact details (email, phone, address) are intentionally omitted from this CV for privacy. Please reach out through LinkedIn or ORCID for academic correspondence.